

Phyla2Vec

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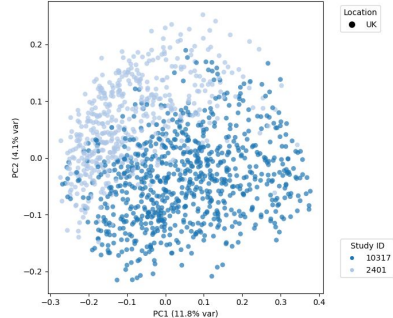
Abstract

- 16S microbiome datasets vary heavily across labs, times, and collection protocols, creating confounding shifts that classifiers must learn
- Classification accuracy is prone to collapsing when test samples use unseen collection methods, even when their underlying microbial content is in-distribution. This makes datasets from different sources incompatible
- We introduce Phyla2Vec, a UniFrac-based [3] encoding that embeds samples into a collection-invariant latent space, enabling robust classification across sources and allowing new samples to be encoded and used immediately without retraining classifiers

Introduction

Batch collection effects introduced per study controlling for confounders age and location. 10317 and 2401 [1] distributions are offset even though both are human stool microbiome.

PCoA - UK-only-human-stool: shape = location, color = study ID



Related Work

Rather than learn representations from abundance tables, our model learns phylogenetic distances between sequences

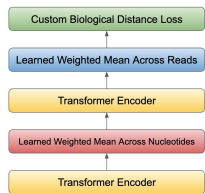
- Microbiome embeddings: Prior works learn representations from abundance tables (DeepMicro autoencoders) or from raw DNA using pretrained transformers (DNABERT) but these approaches lack phylogeny-aware supervision
- Generative models for profiles: Models like MB-GAN simulate realistic abundance tables by matching sparsity, diversity, and co-occurrence structure, but they operate purely in count space. Additionally, diffusion based models like mbVDIT incorporate metadata and tree structure for improved generation and reconstruction quality

Our approach deviates in that our model attempts to learn sequence-level encodings whose geometry directly reflects UniFrac phylogenetic distance, enabling biologically meaningful and collection robust representations

Methodology

Our model uses two transformer encoder blocks with learned weighted averaging layers after each one, combined with a novel UniFrac loss function

Encoding Mechanism



Objective Function

$$\mathcal{L}_2 = \frac{1}{|P|} \sum_{(i,j) \in P} (\alpha d_{ij} + \beta - u_{ij})^2$$

We train our model by re-rarifying each sample (take a new subset of reads) each epoch, and training to minimize the difference of the L2 norm between samples in a batch and their UniFrac distances

Results

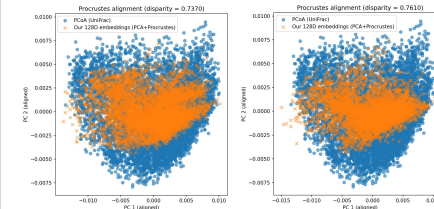
We evaluate our model and each baseline by training a classifier to predict continent of origin on encodings produced from each model. We also classify age using a held-out study to test robustness to collection noise

Model	Location Validation Accuracy
PCoA (UniFrac)	0.9567 ± 0.0075
Transformer Encoder	0.7998 ± 0.0131
MLP Encoder	0.5734 ± 0.0274
VAE	0.6640 ± 0.0171
UniFrac VAE	0.6962 ± 0.0081

Table 1: Validation accuracy (mean ± std) over 6 runs for continent prediction.

Model	Test Accuracy	Validation Accuracy
PCoA (UniFrac)	-	0.7173 ± 0.0097
Transformer Encoder	0.7421 ± 0.0043	0.7433 ± 0.0088
MLP Encoder	0.6749 ± 0.0036	0.6692 ± 0.0041
VAE	0.7142 ± 0.0051	0.7403 ± 0.0060
UniFrac VAE	0.7243 ± 0.0039	0.7388 ± 0.0046

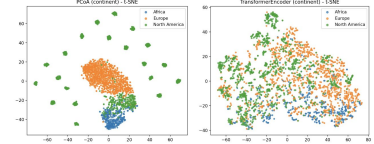
Table 2: Test accuracy for age classification—age ≥ 30—(mean ± std) over 6 runs on held-out study and validation accuracy



Our model produces encodings that are better for classifying age than all other baselines tested, and outperforms all baselines other than PCoA on UniFrac for location classification. Furthermore, our learned embeddings are robust to collection noise, as data in a held-out study can be embedded and classified with similar accuracy to validation data without retraining a classifier

More Results

t-SNE plots demonstrating how learned representations cluster based on the location of the sample compared to UniFrac PCoA



Conclusion

- Built a 16S sequence-to-embedding model that learns a phylogenetically informed latent space using an unweighted UniFrac-based objective.
- Learned weighted pooling [2] and Monte Carlo rarefaction offered modest gains, showing the benefits of learning over attention-enriched embeddings and exposing the model to multiple subsamples
- Main limitation is supervision mismatch: unweighted UniFrac reflects the full ASV table, while the model sees only small stochastic subsets and can often miss rare taxa.
- Despite this gap, the model captures coarse UniFrac structure and provides a foundation for consistent cross-study embeddings and future refinement.

References

- [1] Gonzalez A. et al. Qiita: rapid, web-enabled microbiome meta-analysis. "Nat. Methods" 15, 796–798, 2018.\
- [2] Zichao Y. et al. Hierarchical Attention: Networks for Document Classification. "Association for Computational Linguistics". 2016
- [3] Lozupone C., Knight R. UniFrac: a new phylogenetic method for comparing microbial communities. Applied and Environmental Microbiology 71(12):8228–8235, 2005.